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Socket Client and Socket Server

Program:

Socket_server.c:

```
sahasrasagi@Sahasra: ~/OS_L  ×  +  ∨
GNU nano 8.7.1 socket.
#include <stdio.h>
#include <unistd.h>
#include <sys/socket.h>
#include <sys/un.h>
#include <string.h>

#define SOCKET_PATH "/tmp/my_socket"

int main(){
    int server_fd, client_fd;
    struct sockaddr_un address;
    char buffer[100];

    //create socket
    server_fd = socket(AF_UNIX, SOCK_STREAM, 0);

    //set socket address
    address.sun_family = AF_UNIX;
    strcpy(address.sun_path, SOCKET_PATH);

    // remove old socket
    unlink(SOCKET_PATH);

    //Bind socket to opath
    bind(server_fd, (struct sockaddr *)&address, sizeof(address));

    //wait for client
    listen(server_fd,5);

    printf("Server waiting...\n");

    //accept client
    client_fd = accept(server_fd,NULL,NULL);

    //receive message
    read(client_fd, buffer, sizeof(buffer));

    printf("Message from client: %s\n", buffer);

    close(client_fd);
    close(server_fd);

    unlink(SOCKET_PATH);

    return 0;
```

Socket_client:

```
sahasrasagi@Sahasra: ~/OS_L
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/socket.h>
#include <sys/un.h>
#include <string.h>

#define SOCKET_PATH "tmp/my_socket"

int main(){

    int client_fd;
    struct sockaddr_un address;
    client_fd = socket(AF_UNIX, SOCK_STREAM, 0);

    address.sun_family = AF_UNIX;
    strcpy(address.sun_path, SOCKET_PATH);
    connect(client_fd, (struct sockaddr *)&address,
            sizeof(address));

    write(client_fd, "Hello Server", 12);
    printf("Message sent to server.\n");
    close(client_fd);
    return 0;
}
```

Output:

```
sahasrasagi@Sahasra:~/OS_Lab$ ./socket_server
Server waiting...
Message from client: Hello Server
sahasrasagi@Sahasra:~/OS_Lab$ |
Message sent to server.
sahasrasagi@Sahasra:~/OS_Lab$ ./socket_client
Message sent to server.
sahasrasagi@Sahasra:~/OS_Lab$ |
```